

BENCHMARK EVALUATION: CONVENTIONAL vs. UPGRADED EPS MORTAR

2x2 Quantitative Performance Matrix (Mean ± Standard Deviation)

Key Performance Parameter	Conventional EPS (20% Replacement)	Upgraded EPS (20% Replacement)	Conventional EPS (40% Replacement)	Upgraded EPS (40% Replacement)
28-day Oven-Dry Density (kg/m³)	2011.0 ± 18.5	1969.0 ± 17.0 (Lightweight)	1748.0 ± 15.0	1734.0 ± 14.5 (Lightweight)
Compressive Strength f_c (MPa)	10.15 ± 0.65	15.62 ± 0.85 (+53.9% ▲)	4.20 ± 0.35	5.21 - 7.15 (+70.2% ▲)
Flexural Strength f_r (MPa)	2.10 ± 0.15	3.59 ± 0.20 (+71.0% ▲)	0.95 ± 0.08	1.80 - 1.85 (+94.7% ▲)
Fracture Toughness (mJ/mm³)	12.58 ± 0.85	18.24 ± 1.20 (+45.0% ▲)	1.87 ± 0.15	3.84 ± 0.25 (+105.3% ▲)
Apparent Porosity (%)	7.65 ± 0.25%	6.48 ± 0.24% (Refined)	8.03 ± 0.30%	6.94 ± 0.26% (Refined)
Water Absorption (%)	3.82 ± 0.15%	3.34 ± 0.14% (Reduced)	4.50 ± 0.18%	3.52 ± 0.15% (Reduced)
ITZ Ca/Si Ratio (SEM-EDS)	23.50 ± 2.40 (CH)	6.80 ± 0.55 (C-S-H ▼)	24.85 ± 2.70 (CH)	5.65 ± 0.39 (-77.3% C-S-H ▼)
Strength-to-Cost (MPa/kLKR)	0.275 ± 0.020	0.323 ± 0.025 (+17.5% ▲)	0.098 ± 0.009	0.131 ± 0.011 (+33.7% ▲)
Embodied Carbon (kg CO2/MPa)	53.47 ± 2.80	32.29 ± 1.60 (-39.6% ▼)	130.02 ± 6.50	95.66 ± 4.50 (-26.4% ▼)